

“You Look Yummy!”: Designing a Tangible Interactive Game to Support Cognitive Development in Children with Low-Functioning Autism

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Abstract

Children with low-functioning autism face significant challenges in language and object recognition; however, most existing tools are designed for high-functioning users. We present Yum-Yum, a cognitive training system combining a virtual agent, tangible tools, and game mechanics to support fruit category recognition. Through anthropomorphic prompts and interactive tasks, the system enhances engagement and participation among these children. Preliminary results show that Yum-Yum improves both cognitive ability and interactive behavior among children with low-functioning autism, offering insights for future tools tailored to this group.

CCS Concepts

• **Human-centered computing** → **Accessibility systems and tools**.

Keywords

Social agent; Tangible user interface; Low-functioning autism; Cognitive training

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1 Introduction

Children with low-functioning autism (LFA) often have severe cognitive impairments, repetitive behaviors, and little or no verbal

communication, with some showing aggression or self-injury [1, 6]. Traditional interventions, such as verbal cues, flashcards, or caregiver modeling, often lack intuitiveness and engagement, which makes it difficult to hold the attention of children with LFA or prompt their participation.

Technology-enhanced interventions have proven effective in improving cognition and social skills among children with autism. Various tools, such as emotion recognition apps [4], virtual assistants [2], and joint action systems [3], have shown promise. However, screen-based tools often lack real-world context and are mainly suitable for high-functioning individuals [5]. For children with LFA, who have low literacy and limited attention, these tools’ interactional friendliness and sustained support remain inadequate.

In this research study, we present “Yum-Yum” (Fig. 1a), a tangible user interface (TUI)-based virtual agent game designed to support cognitive training for children with LFA. This game aims to enhance children’s ability to recognize and identify specific objects (e.g., fruits). We created a virtual environment integrated with interactive projection technologies, allowing children to engage in both verbal and nonverbal interactions with a three-dimensional (3D) virtual agent embedded in a realistic social context using familiar physical objects. The virtual agent detects and responds to real-world objects in real time through sensors, enabling bi-directional human-agent interactions. By bridging the gap between physical and virtual spaces, the Yum-Yum system promotes more natural behaviors and improves children’s attention and engagement.

2 System Design

Yum-Yum is a single-player game based on interactive projection technology. The gameplay centers around a 7-year-old virtual agent named Eric and a TUI-supported virtual social environment. The Yum-Yum system supports multimodal interaction, including human-agent dialogues and tangible manipulations that link physical objects with virtual projections. Children are guided through three task-oriented mini-games via social interaction with Eric and they ultimately feed three types of fruit (i.e., apples, bananas, and watermelons) to the corresponding animals.

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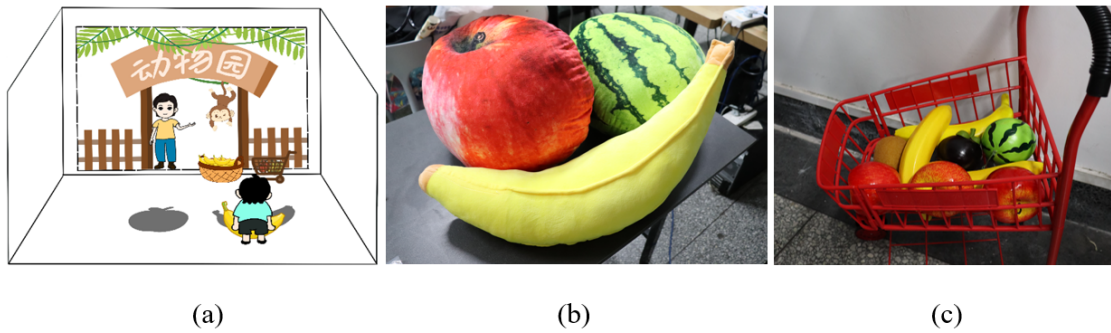


Figure 1: (a) A hand-drawn overview of the Yum-Yum system; (b) fruit cushions from Game 1 and (c) fruit props and cart from Game 2.

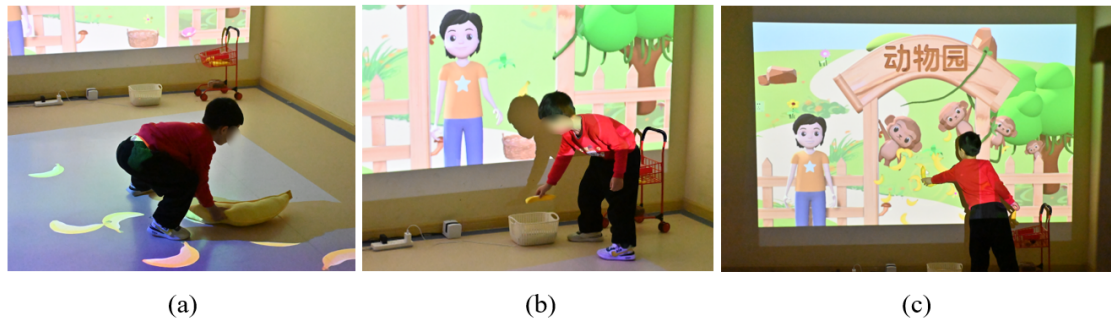


Figure 2: Mini-game interfaces for (a) Fruit Recognition, (b) Fruit Classification, and (c) Feeding Animals.

2.1 Fruit Recognition

In Game 1, children pick up physical fruit toys, while the floor projection displays 3D models of three virtual fruits. Following Eric’s verbal prompts, they place the correct fruits in the designated locations.

2.2 Fruit Classification

In Game 2, children search for the target fruits among various distractors on a rolling cart and place the correct item into a real fruit basket. Completion of the task advances them to the next level.

2.3 Feeding Animals

In Game 3, a virtual zoo scene is projected onto the wall. The children can directly interact with the corresponding fruit images by tapping on them following Eric’s guidance to feed the zoo animals.

To match the children’s natural interaction habits, we used familiar fruit toys as tangible interfaces, combined with verbal and nonverbal feedback from the virtual agent to guide their engagement. Visual and auditory rewards were used to promote participation across the three tasks. For example, correctly placing a fruit triggered celebratory fireworks using animations and sounds. Upon completing all tasks, a message like “Great job, my friend!” was shown to the children to reinforce positive feedback and intrinsic motivation.

3 Technical Implementation

We developed the Yum-Yum platform using Unity3D (Unity Technologies, San Francisco, CA, USA), with two projectors displaying game scenes on the floor and wall. The Yum-Yum system features

three task-based mini-games with distinct technical implementations. In Game 1, a teacher interface verifies whether the child correctly places the fruit-shaped cushion to trigger game interactions. Game 2 employs radio-frequency identification (RFID) to detect the correct fruit placement automatically, removing the need for teacher input. Game 3 uses light detection and ranging (LiDAR) to enable touch interaction on the wall, allowing children to tap virtual fruits to complete their feeding tasks.

4 Evaluation

We conducted a pilot study at an autism rehabilitation center, where we recruited 10 children with autism (T1-T10, 8 boys, 2 girls, mean = 5.3, standard deviation = 1.06). The ethics committee of our university approved this study and written informed consent was obtained from all parents.

We adopted an exploratory approach to evaluate the usability of the Yum-Yum system and children’s engagement levels. All participants successfully completed the three mini-games. Notably, three children completed Game 2 and four children completed Game 3 entirely independently under the virtual agent’s guidance. Some children initially experienced difficulties in Game 1, likely due to their uncertainty when interacting with unfamiliar technology. However, most children exhibited strong interest in the virtual agent, Eric. For instance, T3 frequently touched and examined Eric’s projected image, regarding it as a “real social partner.” The children were generally enthusiastic about using tangible props to interact with the virtual scene. T1, T2, and T9 held the fruit-shaped pillows throughout the session, while T6 and T8 pretended to “eat”

the fruit toys. According to the therapists, the plush materials provided tactile comfort and a sense of security. Despite their language impairments, most children attempted verbal responses using simple words or sounds and initiated nonverbal behaviors, such as touching, hugging, or offering objects to Eric, which demonstrated unexpected social initiative. Overall, the results indicate that the Yum-Yum system shows promising potential for enhancing cognitive skills, task focus, and social interaction among children with autism.

5 Conclusion and Future Work

In summary, the Yum-Yum system effectively supported full-body interaction, enhancing cognitive and social skills among the children with LFA. Future work will expand Eric's dialogue corpus and improve the naturalness of its verbal and nonverbal responses to better meet these children's cognitive needs. We will also refine the Yum-Yum system by adding more cognitive items and testing its adaptability in diverse real-world contexts.

Acknowledgments

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